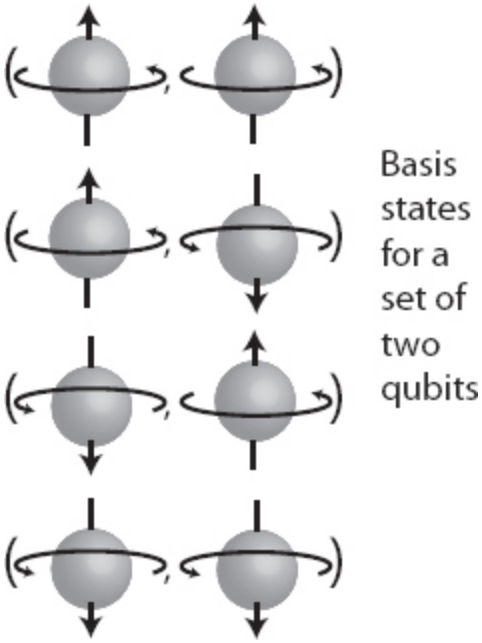
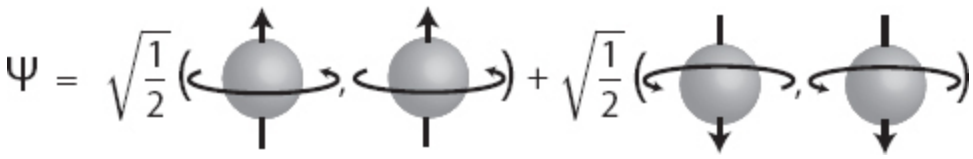


Just because we have two qubits, it doesn't mean they are necessarily entangled. Consider a state that is simply one of the basis states, say, the one where both qubits are spin-up. If Alice measures her qubit along the vertical axis, she will always obtain spin-up, and likewise for Bob. If Alice measures her spin along the horizontal axis, she has a fifty-fifty chance of getting spin-right or spin-left, and again likewise for Bob. But in each case, we don't learn anything about what Bob will see by learning what Alice saw. That's why we can often casually speak of "the wave function of a particle," even though we know better—when different parts of the system are unentangled with each other, it's just as if they have their own wave functions.

Instead, let's consider an equal superposition of two basis states, one with both spins up, and the other with both spins down:



If Alice measures her spin along the vertical axis, she has a fifty-fifty chance of getting spin-up or spin-down, and likewise for Bob. The difference now is that if we learn Alice's outcome before Bob does his measurement, we know what Bob will see with 100 percent confidence—